Paper 27 - Observer Delay and Shared Reality

CQE-paper-27

CQECMPLX Formal Paper Series

Review-facing PDF built 2026-06-13

Construction Basis

Explore observer delay, sampling buffers, and shared-state constraints as an interpretation layer beyond the proof stack.

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-27.md
- paper kernel manifest: production\paper-kernels\papers\CQE-paper-27\paper_kernel_manifest.json
- verifier: production\formal-papers\CQE-paper-27\verify_observer_delay_shared_reality.py
- receipt: production\formal-papers\CQE-paper-27\observer_delay_shared_reality_receipt.json (CQE-paper-27: pass_with_interpretive_obligations)
- body: production\papers\CQE-paper-27\PAPER-BODY.md
- source: production\papers\CQE-paper-27\SOURCE.md
- formal: production\papers\CQE-paper-27\01-CQE-formal\FORMAL.md
- tool: production\papers\CQE-paper-27\02-CQE-tool\TOOL.md
- workbook: production\papers\CQE-paper-27\03-CQE-workbook\WORKBOOK.md
- recovered_output: production\lib-forge\recovered\papers_output\CQE-paper-27.md

Paper 27 - Observer Delay and Shared Reality

Abstract

Paper 27 closes a finite observer-frame statement and preserves the human observer language as interpretation. The closed result is not consciousness, measurement collapse, or relativistic simultaneity. It is a chart theorem: the static four-frame Z4 template is exact; opposite-boundary reads share the same center coordinate `C`; read-then-anneal delay is bounded by the finite chart; and the tempting promotion from static Z4 to temporal Rule 30 period is explicitly refuted over the tested trace.

The formal verifier returns `pass_with_interpretive_obligations`. It verifies the package's static Z4 template, calls the temporal-scope checker through depth 512, walks 64 observer-window rows, and records all interpretation claims as obligations rather than proof.

Closed Evidence

The receipt is generated by `production/formal-papers/CQE-paper-27/verify_observer_delay_shared_reality.py`. It imports the live `centroid_voa`, `temporal_z4_scope`, and Rule 30 canonical row APIs.

The static Z4 verifier passes with two fixed points, zero period-2 states, and six period-4 states. The fixed points are `(0,0,0)` and `(1,1,1)`, each with four-frame label `(0,0,0,0)`. The other six chart states have period 4.

The temporal verifier returns `static_template_only`. Over the tested depth window of 512, neither the four-frame label trace nor the Rule 30 center column is periodic at periods 1, 2, or 4. The first label counterexamples occur at indices 1, 2, and 6 for periods 1, 2, and 4 respectively. The first center-bit counterexamples occur at indices 1, 3, and 6.

The shared-center receipt walks 64 Rule 30 observer rows. Every opposite- boundary pair preserves the same `C`. The rows still record boundary-side disagreement 37 times, so agreement on `C` does not erase the boundary data. The anneal-delay distribution is 27 rows at delay 0, 20 rows at delay 2, and 17 rows at delay 3. The maximum delay is 3.

Definitions

An observer window is a local state `(L, C, R)` read from a Rule 30 predecessor row. The two boundary observers read the same center from opposite sides: `(L, C, R)` and `(R, C, L)`.

The shared center is `gluon(s) = C`. It is invariant under the LR-podal reflection and is the only coordinate this paper promotes as a shared finite fact.

The four-frame label is the tuple of wrap-step counts in the four centroid frames: C-centroid, R-centroid, C-flipped, and L-centroid. Its Z4 period is a static label period, not a time period.

Observer delay is the finite chart lag between reading a window and committing the result of `anneal_to_lie_conjugate`. This paper measures delay in S3 anneal steps. It does not measure human latency.

Shared reality is an interpretive phrase for agreement on `C`. The formal object is only the shared-center receipt.

Claims

- The static four-frame Z4 template is exact over the eight chart states.
- The static Z4 template does not promote to a temporal Rule 30 period over the tested trace.
- Opposite-boundary observers agree on `C` for the same local window.
- Boundary disagreement remains visible as data; agreement on `C` does not delete `L/R` residue.
- Read-then-anneal delay is bounded by at most three finite chart steps.
- Consciousness, collapse, relativistic simultaneity, and human-latency readings are not closed claims of this paper.

Theorem 27

Observer delay and shared reality are admissible CQE terms only when reduced to finite receipts: static frame labels, shared-center equality, bounded anneal delay, and explicit temporal-period refutation. Any psychological, physical, or metaphysical observer reading remains an obligation.

Proof

The static frame result follows from `verify_z4_period_template`. The verifier reports exactly two fixed points, zero period-2 states, and six period-4 states. Therefore the Z4 label template is a real finite chart object.

The temporal boundary follows from `verify_temporal_z4_scope(max_depth=512)`. That function computes predecessor-state four-frame labels and center-column bits along the actual Rule 30 trace, then tests periods 1, 2, and 4. It returns `static_template_only`, with counterexamples for each tested period in both the label trace and the center column. Therefore the static Z4 template is not a temporal sampling period in the tested trace.

The shared-center result follows from `verify_gluon_invariance` and the 64-row observer receipt. For every row, the opposite-boundary state swaps `L` and `R` while leaving `C` unchanged. The receipt confirms that all rows share the same center. It also records 37 side-disagreement rows, proving that the receipt preserves boundary residue rather than flattening the read.

The delay bound follows from `anneal_to_lie_conjugate`. Every sampled row commits to a Lie-conjugate attractor in at most three S3 transposition steps. The observed distribution is 27 rows at delay 0, 20 at delay 2, and 17 at delay 3. This proves a bounded finite-chart delay and nothing stronger.

Worked Example

Take the first 64 Rule 30 observer windows. Two observers read each window from opposite boundary directions. Each row records the original state, the reflected state, both center reads, whether the sides disagree, the anneal delay, the four-frame label, the static Z4 period, and the emitted center bit.

The solved result is a mixed receipt: all rows agree on `C`; many rows disagree on side; delay is bounded; temporal Z4 periodicity is refuted. That is the proper scientific shape of this paper. The observer vocabulary is useful only because the receipt keeps the distinction visible.

Open Obligations

Human latency is not claimed. Finite anneal steps are not measured response times.

Shared reality is interpretive. The formal result is agreement on `C`, not a proof of physical simultaneity.

Consciousness and measurement collapse are not claimed. Static frame labels and center invariance do not imply either one.

Temporal Z4 period is refuted for the tested trace. Any future claim of a temporal period must overcome the recorded counterexamples.

Suite Role

Paper 27 gives the suite a disciplined observer vocabulary. It allows later papers to talk about a selected center, a delay before commitment, and a shared invariant, while preventing those words from silently becoming physics or consciousness claims. The paper's most important contribution is the boundary: static structure is not automatically temporal structure.